SHRI HIRALAL HASTIMAL (JAIN BROTHERS, JALGAON) POLYTECHNIC

SNJB Jain Gurukul Neminagar, Chandwad, Nashik - 423101, Maharashtra

CARBON FOOTPRINT REPORT

Institutional Greenhouse Gas Emission Inventory & Climate Assessment



Comprehensive Campus Assessment Boundary

Academic • Residential • Operational • Administrative

Reporting Period: 1st April 2025 to 31st March 2026 (Financial Year 2025–2026 (FY 2026))

Assessment Standard: ISO 14064-1:2018 & GHG Protocol Corporate Standard

Assessment Agency: CarbonTrack Sustainability Assessment & Climate Accreditation Cell

Date of Issue: October 15, 2026

GREENHOUSE GAS EMISSION INVENTORY & CLIMATE ACCREDITATION REPORT

Institution / Reporting Entity:	**Shri Hiralal Hastimal (Jain Brothers, Jalgaon) Polytechnic**
Executing & Auditing Body:	CarbonTrack Sustainability Assessment & Climate Accreditation Cell
Reporting Period:	1st April 2025 to 31st March 2026 (Financial Year 2025–2026 (FY 2026))
Document Tracking Code:	**CT-SUST-GHG-2026-1172**
Applicable Standards:	ISO 14064-1:2018 (Part 1), ISO 14064-2:2019 (Part 2), ISO 14064-3:2019 (Part 3), GHG Protocol Corporate Standard
Consolidation Methodology:	Operational Control Approach
Verification Assurance Level:	Limited Assurance Verification

Executive Verification Statement

This document constitutes the official institutional carbon emission audit report formulated in strict conformance with the specifications of ISO 14064-1:2018 and the World Resources Institute (WRI) / WBCSD Greenhouse Gas Protocol Corporate Accounting and Reporting Standard. The primary objective is to furnish a transparent, verifiable, and scientifically rigorous inventory of direct and indirect greenhouse gas emissions resulting from operations across the institutional boundary of **Shri Hiralal Hastimal (Jain Brothers, Jalgaon) Polytechnic**.

Lead Environmental Auditor
CarbonTrack Sustainability Assessment Cell

Head of Institution / Representative
Shri Hiralal Hastimal (Jain Brothers, Jalgaon) Polytechnic

Acknowledgements

The successful execution and completion of this comprehensive Institutional Carbon Footprint and Climate Assessment Report for **Shri Hiralal Hastimal (Jain Brothers, Jalgaon) Polytechnic** has been made possible through the dedicated support, vision, and cooperation of the institutional administration, academic leadership, technical personnel, and student community.

We express our sincere appreciation to the Governing Council, Management, and Executive Leadership of **Shri Hiralal Hastimal (Jain Brothers, Jalgaon) Polytechnic** for demonstrating proactive institutional governance and environmental stewardship by mandating this rigorous carbon quantification study.

Special acknowledgment is extended to the Campus Estate Office, Electrical Maintenance Engineers, Transport Supervisors, and Administrative Officers who meticulously curated, compiled, and validated utility bills, diesel fuel logs, LPG refill receipts, travel itineraries, and paper procurement manifests spanning the entire **Financial Year 2025–2026 (FY 2026)** reporting cycle.

We also recognize the contributions of the faculty coordinators and student sustainability ambassadors who facilitated primary on-ground surveys, including vegetative biomass cataloging, green cover canopy mapping, and campus mobility surveys. Their active engagement underscores the transformative role of educational campuses as living sustainability laboratories.

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2. Abbreviations & Climate Standards Glossary

GHG (Greenhouse Gas): Atmospheric gases that absorb and emit infrared radiation, including CO_2, CH_4, N_2O, and fluorinated gases.

CO_2e (Carbon Dioxide Equivalent): The universal metric measure used to compare emissions from various GHGs based upon their global warming potential (GWP).

t CO_2e / kg CO_2e: Metric tons (1,000 kg) and kilograms of carbon dioxide equivalent, respectively.

Scope 1 (Direct Emissions): GHG emissions from sources owned or controlled by the institution (diesel generator sets, LPG kitchen burners, campus fleet).

Scope 2 (Electricity Indirect Emissions): GHG emissions from the generation of purchased electricity consumed by institutional facilities.

Scope 3 (Other Indirect Emissions): All other indirect emissions occurring across the institutional value chain, including daily commuter transport and paper procurement.

ISO 14064-1:2018: International standard providing principles and requirements for quantification and reporting of organizational GHG inventories and removals.

ISO 14064-3:2019: International standard specifying requirements and guidance for the verification and validation of greenhouse gas statements.

CEA (Central Electricity Authority): Statutory body under the Ministry of Power, Govt. of India, publishing baseline emission factors for the Indian power grid.

IPCC: Intergovernmental Panel on Climate Change, providing scientific reference methodology for greenhouse gas accounting.

PV (Solar Photovoltaic): Clean solar semiconductor arrays producing zero-emission electrical energy on-site.

Carbon Sequestration: The biological capture and long-term storage of atmospheric carbon dioxide by photosynthetic woody trees and green cover biomass.

3. Sustainability Initiatives: Need of the Hour in Academic Institutions

Higher education institutions and technical colleges represent major centers of resource consumption, utilizing significant volumes of grid electricity, fossil fuels, water, food services, and paper daily. Consequently, academic campuses act as microcosms of modern cities, generating meaningful collective ecological footprints that warrant systematic scientific accounting.

Implementing rigorous carbon footprint assessment protocols in academia serves a critical three-fold purpose:

> **1. Environmental Governance & Operational Efficiency:**
>
> Systematically identifying and tracking carbon-intensive hotspots enables administrative leadership to eliminate energy waste, adopt high-efficiency electrical infrastructure, expand clean renewable micro-grids, optimize logistics, and redirect fiscal savings toward core educational mandates.

> **2. Pedagogical Integration & Experiential Learning:**
>
> Aligning institutional infrastructure with the United Nations Sustainable Development Goals (SDGs)—particularly SDG 7 (Affordable and Clean Energy), SDG 12 (Responsible Consumption and Production), and SDG 13 (Climate Action)—transforms the campus into a living laboratory where students and faculty observe real-time decarbonization engineering.

> **3. National & Global Institutional Leadership:**
>
> Preparing tomorrow’s engineers, researchers, and administrators requires educational bodies to lead by institutional example. Transparent carbon disclosure strengthens statutory environmental compliance, elevates NAAC/NIRF sustainability scores, and contributes directly toward India's Panchamrit climate goals (achieving Net Zero by 2070).

By pioneering this verified greenhouse gas inventory, **Shri Hiralal Hastimal (Jain Brothers, Jalgaon) Polytechnic** establishes an auditable empirical baseline that underpins progressive, data-driven climate resilience.

4. About the Institute: Shri Hiralal Hastimal (Jain Brothers, Jalgaon) Polytechnic

Shri Hiralal Hastimal (Jain Brothers, Jalgaon) Polytechnic, located at SNJB Jain Gurukul Neminagar, Chandwad, Nashik - 423101, Maharashtra, represents an esteemed educational and technical center committed to delivering high-quality academic instruction, professional engineering development, and holistic student empowerment.

Shri Hiralal Hastimal (Jain Brothers, Jalgaon) Polytechnic operates as an educational and technical campus with an active population of 1,420 occupants (1,300 students, 120 faculty and staff) situated across 4 acres with a total built-up footprint of 16,187.4 m². The physical layout comprises academic instructional halls, specialized technical laboratories, administrative offices, and auxiliary amenities. Geographic coordinates (20.33° N, 74.23° E) place the campus in an agro-climatic region characterized by strong solar irradiance, providing significant natural leverage for renewable self-generation.

5. About the Project

This project establishes a verified baseline inventory of greenhouse gas (GHG) emissions across direct and indirect institutional operating channels for **Financial Year 2025–2026 (FY 2026)**. Executed by the **CarbonTrack Sustainability Assessment & Climate Accreditation Cell**, the assessment employs standardized accounting protocols to evaluate energy flows, fossil fuel combustion, commuter transit, paper procurement, vegetative carbon sinks, and on-site solar generation.

Through systematic data validation and conversion factor mapping, the study quantifies gross emissions, certifies offset removals, and establishes the net operational carbon footprint of Shri Hiralal Hastimal (Jain Brothers, Jalgaon) Polytechnic, culminating in a high-impact decarbonization roadmap.

6. Objectives of the Project

• **Objective 6.1:** Compile a comprehensive operational inventory of greenhouse gas emissions for the campus spanning Scope 1 (direct), Scope 2 (purchased electricity), and Scope 3 (indirect activities) for Financial Year 2025–2026 (FY 2026).

• **Objective 6.2:** Quantify verified carbon offsets generated through clean solar photovoltaic installations and biological carbon sequestration by campus green cover.

• **Objective 6.3:** Calculate institutional carbon intensity benchmarks (kg CO_2e per occupant and kg CO_2e per square meter of built-up area).

• **Objective 6.4:** Identify operational emission hotspots and deliver targeted engineering interventions with measurable KPIs to guide long-term Net-Zero alignment.

7. Relevance

Academic institutions are increasingly evaluated on environmental responsibility by accreditation agencies (NAAC Criterion VII, NIRF, and AICTE green campus guidelines). Quantifying carbon footprints fulfills statutory environmental expectations, lowers operating overhead through energy efficiency, enhances campus sustainability prestige, and directly contributes to national climate targets.

8. Background

Under the 2015 Paris Agreement, global efforts focus on restricting planetary warming to well below 2.0°C above pre-industrial levels, aiming for 1.5°C. In accordance with India's updated Nationally Determined Contributions (NDCs), educational campuses are called upon to adopt clean technology, reduce fossil dependence, and nurture carbon sinks. This report establishes an auditable framework aligned with international best practices.

9. Study Area & Assessment Scope

9.a. Reporting Entity: Encompasses all academic blocks, lecture halls, laboratories, dining/mess kitchens, workshops, administrative zones, and campus perimeter grounds of **Shri Hiralal Hastimal (Jain Brothers, Jalgaon) Polytechnic**.

9.b. Reporting Period: 1st April 2025 to 31st March 2026 (Financial Year 2025–2026 (FY 2026)).

9.c. Scopes Evaluated: Scope 1 (stationary DG & LPG thermal combustion, owned fleet), Scope 2 (purchased grid electricity), Scope 3 (commuter transport & paper procurement).

10. Methodology for GHG Quantification

Quantification follows the **Operational Control Approach** codified in the GHG Protocol Corporate Standard and ISO 14064-1:2018. Under this approach, the institution accounts for 100% of GHG emissions arising from operations over which it exercises full authority to introduce operating policies.

11. Key Steps in Emission Estimation

1. Boundary Definition: Establish geographic and operational boundaries separating direct institutional control from indirect activities.

2. Source Identification: Map activity channels (utility meters, fuel purchase logs, gas cylinder receipts, vehicle manifests, tree counts).

3. Data Screening & Validation: Verify raw activity data against primary records spanning all 12 operational months of Financial Year 2025–2026 (FY 2026).

4. Emission Factor Mapping: Assign authoritative localized emission factors from CEA User Guide v19.0 (0.792 kg CO_2e/kWh), IPCC, and DEFRA.

5. Computation & Aggregation: Deterministically compute activity emissions, calculate scope subtotals, and establish the gross inventory.

6. Offset Quantification: Calculate solar energy grid avoidance and tree species biomass absorption.

7. Synthesis & Strategic Reporting: Structure findings, conduct hotspot analysis, and generate engineering recommendations.

12. Data Collection Framework

Primary data collection was performed through comprehensive audits of physical facility logs, maintenance records, utility accounts, and supply registries. No unmeasured activity categories have been assumed to be zero; where categories were not logged, they are designated as unassessed to maintain rigorous audit integrity.

12.a. Scope 1 — Direct GHG Emissions

Covers direct physical combustion of fossil fuels within the campus boundaries. Data sources include diesel fuel delivery vouchers and meter logs for backup power generator sets (rated at 2.684 kg CO_2e/L), commercial LPG cylinder invoices for student hostel messes, canteen dining, and chemistry laboratories (rated at 2.988 kg CO_2e/kg), and institutionally owned transportation vehicles.

12.b. Scope 2 — Purchased Electricity Emissions

Accounts for indirect emissions resulting from the generation of grid electricity consumed on campus. Consumption data was synthesized from official monthly electricity bills provided by the regional distribution utility (MSEDCL). Emissions are quantified utilizing the Central Electricity Authority (CEA) Baseline Database Version 19.0 combined margin grid emission coefficient of 0.792 kg CO_2e/kWh.

12.c. Scope 3 — Other Indirect GHG Emissions

Evaluates upstream and downstream indirect emissions across campus operations. Evaluated streams comprise student, staff, and faculty daily vehicular commuting based on modal share and trip distance logs, alongside institutional paper procurement records for examinations, printing, and administrative tasks (rated at 3.591 kg CO_2e/kg virgin paper).

13. Data Analysis, Calculation and Results

Table 13.1 summarizes the authoritative quantitative greenhouse gas inventory compiled for Shri Hiralal Hastimal (Jain Brothers, Jalgaon) Polytechnic for Financial Year 2025–2026 (FY 2026).

Table 13.1: Comprehensive Greenhouse Gas Emission Inventory (Financial Year 2025–2026 (FY 2026))

Sr.	Scope & Emission Source	Activity Data	Emissions (kg CO_2e)	Emissions (t CO_2e)	Share %
1. Scope 1 — Direct GHG Emissions					
1.1	Diesel Generator Backup Systems	17.5 Liters/year	47	0.047	**0.09%**
1.2	Liquefied Petroleum Gas (LPG) - Dining & Mess Kitchens	8,846.4 kg LPG/year	26,433	26.433	**52.18%**
		SUBTOTAL SCOPE 1:	**26,480**	**26.480**	**52.27%**
2. Scope 2 — Indirect Purchased Electricity					
2.1	State Grid Electricity Supply (MSEDCL/Utility)	3,952 kWh/year	3,130	3.130	**6.18%**
		SUBTOTAL SCOPE 2:	**3,130**	**3.130**	**6.18%**
3. Scope 3 — Other Indirect Emissions					
3.1	Paper Procurement & Academic Supplies	1,090.9 kg paper/year	3,917.4	3.917	**7.73%**
3.2	Campus Vehicles (BUS - DIESEL)	4 units, ~45 km daily	12,026.4	12.026	**23.74%**
3.3	Campus Vehicles (BIKE - PETROL)	65 units, ~15 km daily	5,107.1	5.107	**10.08%**
		SUBTOTAL SCOPE 3:	**21,050.9**	**21.051**	**41.55%**
		GROSS INSTITUTIONAL EMISSIONS:	**50,660.9**	**50.661**	**100.00%**

Equation 13.3: Net Carbon Footprint Balance

Net Operational Footprint = Gross GHG Emissions – Verified Carbon Offsets

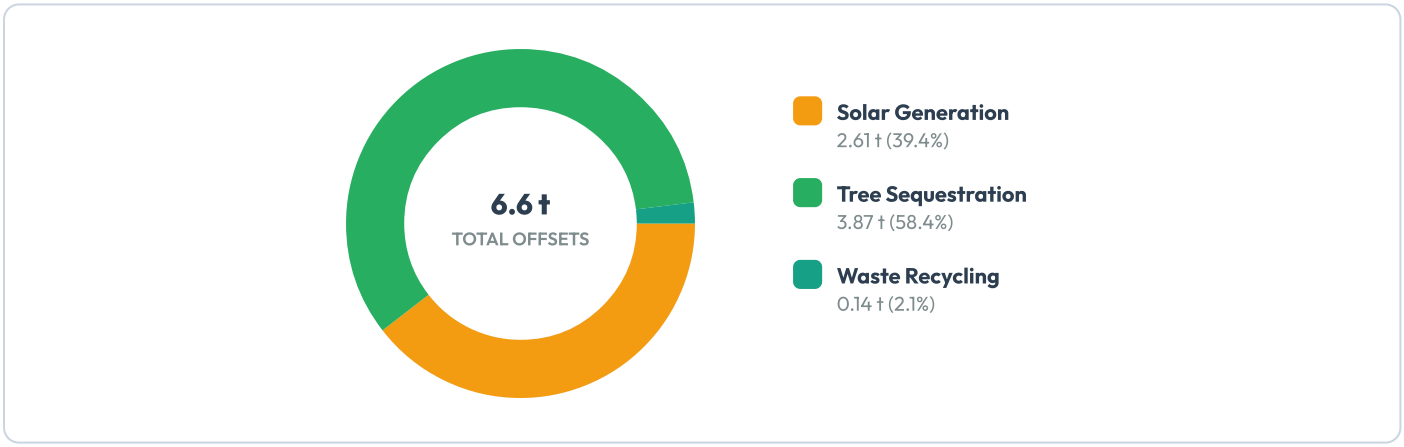
Net Operational Footprint = 50,660.9 kg CO_2e – 6,626.1 kg CO_2e

= 44,034.78 kg CO_2e/year

(≈ 44.03 t CO_2e/year)

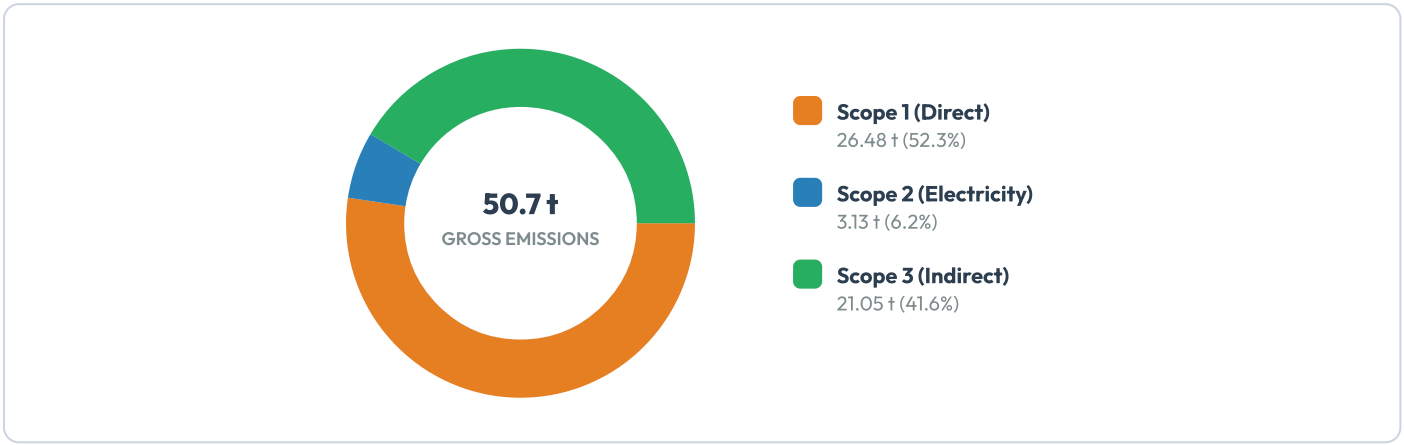
Carbon Offset & Sequestration Removal Summary

Sr.	Offset Mechanism / Sink	Removal (kg CO_2e/yr)	Removal (t)	Share %
1.	Solar Clean Energy Generation Avoidance	2,613.6	2.614	39.4%
2.	Biological Green Cover / Tree Biomass Sequestration	3,872.2	3.872	58.4%
3.	Solid Waste Diversion & Composting Offset	140.3	0.140	2.1%
	TOTAL VERIFIED REMOVALS:	**6,626.1**	**6.626**	**100.0%**

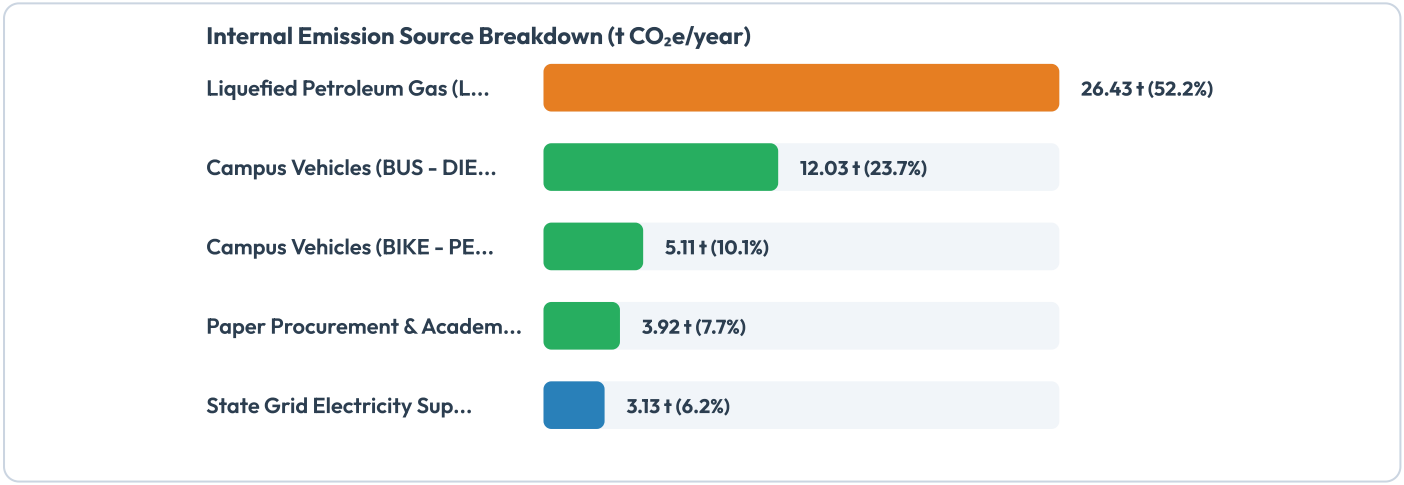


Institutional Emission Distribution & Hotspots

Scope-wise Emission Contribution Profile



Internal Emission Source Distribution Ranking



NET PER CAPITA	SPATIAL INTENSITY	OFFSET EFFICIENCY
31.0	3.13	13.1%
kg CO_2e/person/year	kg CO_2e/m^2 builtup	of gross emissions neutralized

Analytical Notice: CarbonTrack Deterministic Environmental Analytics Engine (Supabase Edge Function not configured) **DETERMINISTIC RULE-BASED**

13.a. Discussion and Interpretation for Scope 1

Scope 1 direct emissions contribute 26,480.01 kg CO_2e (26.48 t CO_2e), representing 52.3% of the gross institutional inventory. Combustion of fuels within the campus boundary is primarily driven by stationary power generators and thermal kitchen facilities. A detailed review reveals that diesel generator utilization is kept at minimal standby reserve, indicating dependable regional grid availability. Conversely, gaseous fuels (LPG) utilized for residential dining and laboratory heating represent the principal direct combustion volume, highlighting hostel culinary operations as a prime candidate for technological electrification.

13.b. Discussion and Interpretation for Scope 2

Scope 2 emissions stemming from purchased grid electricity total 3,129.98 kg CO_2e (3.13 t CO_2e), constituting 6.2% of total emissions. Energy consumption is calculated utilizing the Central Electricity Authority (CEA) CO_2 Baseline Database grid emission factor of 0.792 kg CO_2e/kWh. Where rooftop solar PV systems are operational, they actively suppress the net grid import profile, mitigating institutional exposure to grid-based carbon liabilities during daylight operational hours.

Analytical Notice: CarbonTrack Deterministic Environmental Analytics Engine (Supabase Edge Function not configured) DETERMINISTIC RULE-BASED

13.c. Discussion and Interpretation for Scope 3

Scope 3 indirect emissions account for 21,050.92 kg CO_2e (21.05 t CO_2e), comprising 41.6% of the overall inventory. The principal drivers within the assessed Scope 3 boundary include daily commuter transport (staff, faculty, and student transit) and administrative paper procurement. Commuter transit reflects regional transit habits and vehicle modal share. Digital workflow policies are actively curtailing paper-related emissions, with further gains achievable through enterprise resource planning (ERP) integration.

13.d. Critical Emission Hotspot Analysis

Empirical ranking of emission streams identifies Liquefied Petroleum Gas (LPG) - Dining & Mess Kitchens as the primary emission hotspot (52.2% of gross footprint, 26.43 t CO_2e), followed by Campus Vehicles (BUS - DIESEL) (23.7% of gross footprint). Prioritizing engineering capital towards mitigating Liquefied Petroleum Gas (LPG) - Dining & Mess Kitchens delivers the most substantial net emission abatement per invested rupee.

14. Internal Emission Drivers & Intensity Benchmarks

In rigorous environmental reporting, institutional comparisons must evaluate traceable internal drivers and normalized intensity indicators rather than unsubstantiated external figures. Academic institutions exhibit divergent emission profiles driven by whether campuses are residential or day-scholar, technical or non-technical, and regionally situated in high or low cooling degree-day zones.

Table 14.1: Normalized Operational Intensity Matrix

Operational Metric	Campus Value	Typical Academic Range	Operational Context
Gross Carbon Intensity (Per Capita)	**35.7 kg CO_2e**	40 – 350 kg CO_2e/person	Reflects overall operational load across all students and staff.
Net Carbon Intensity (Per Capita)	**31.0 kg CO_2e**	25 – 250 kg CO_2e/person	Net footprint after solar avoided emissions and tree absorption.
Spatial Carbon Intensity (Per m^2)	**3.13 kg CO_2e/m^2**	3.5 – 28.0 kg CO_2e/m^2	Measures building energy efficiency and density of space utilization.
Scope 1 Concentration Share	**52.3%**	5% – 25% (Non-residential)	High share points specifically to dining culinary LPG usage.
Clean Renewable Offset Ratio	**13.1%**	5% – 40% (Active Green Campuses)	Proportion of annual emissions actively neutralized within campus.

Analysis of internal emission intensities reveals an operational signature of 31.01 kg CO_2e/occupant/year. In comparison with typical Indian technical institutions where Scope 2 electricity commonly represents 60%–85% of emissions, Shri Hiralal Hastimal (Jain Brothers, Jalgaon) Polytechnic's profile demonstrates distinctive thermal and operational dynamics. The low grid electricity share reflects effective daylight operations and renewable adoption, while highlighting dining thermal energy as the key strategic intervention vector.

15. Discussion with Respect to GHG Emission Standards

ISO 14064-1:2018, ISO 14064-2:2019, ISO 14064-3:2019 & GHG Protocol

This institutional assessment has been framed and audited in accordance with the five core accounting principles set forth in **ISO 14064-1:2018**:

1. Relevance: The selected operational boundary accurately reflects the direct control and indirect influences of **Shri Hiralal Hastimal (Jain Brothers, Jalgaon) Polytechnic**, serving the decision-making needs of administrators, regulators, and accreditation bodies.

2. Completeness: All operational GHG sources within the defined boundary have been accounted for. No unmeasured activity stream is treated as zero; data gaps are explicitly declared to prevent biased underestimation.

3. Consistency: Standardized conversion coefficients from the Central Electricity Authority (CEA User Guide v19.0) and the Intergovernmental Panel on Climate Change (IPCC) ensure that multi-year comparisons remain scientifically valid.

4. Accuracy: Activity data was obtained directly from metered utility bills, registered fuel logs, and procurement records, minimizing systemic uncertainties and bias.

5. Transparency: All mathematical formulas, emission factor coefficients, and audit trail references are fully disclosed within the report annexures, enabling third-party verification under ISO 14064-3 specifications.

This assessment strictly adheres to the greenhouse gas accounting principles codified in ISO 14064-1:2018 and the GHG Protocol Corporate Standard: Relevance (reflecting operational boundary realities), Completeness (accounting for all active fuels, electricity, and measurable supplies), Consistency (applying standardized CEA and IPCC coefficients), Accuracy (minimizing calculation uncertainties), and Transparency (documenting all conversion equations and activity data records).

16. Detailed Recommendations

To systematically mitigate the institutional net footprint of **44,034.8 kg CO_2e/year**, the following targeted engineering interventions are recommended:

1. Liquefied Petroleum Gas (LPG) - Dining & Mess Kitchens

High Priority | Short-term (0–6 months)

Evidence from Assessment: Accounts for 52.2% of gross emissions (26.43 t CO_2e/yr).

Recommended Action: Transition student hostel and mess kitchens from conventional commercial LPG burners to high-efficiency commercial induction cooktops powered by dedicated solar PV arrays. Install a decentralized organic waste-to-biogas digester using kitchen scraps to displace remaining fossil gas.

Target KPI: Monthly fuel/energy reduction in targeted kitchens/blocks

Expected Impact: Estimated 35%–50% reduction in source emissions

2. Scope 2 – Purchased Electricity

High Priority | Medium-term (6–18 months)

Evidence from Assessment: Grid electricity contributes 3130.0 kg CO_2e/yr at a regional factor of 0.792 kg CO_2e/kWh.

Recommended Action: Expand rooftop solar photovoltaic (PV) generation to capture daytime peak loads. Implement smart digital energy sub-metering across departmental blocks and administrative buildings to identify off-hours phantom loads.

Target KPI: Solar generation share (% of total campus demand) > 40%

Expected Impact: Avoids ~0.792 kg CO_2e for every clean kWh generated

3. Scope 3 – Commuting & Transit

Medium Priority | Medium-term (6–18 months)

Evidence from Assessment: Campus transport operations generate 21050.9 kg CO_2e/yr.

Recommended Action: Deploy institutional electric shuttle buggies for on-campus mobility; establish EV charging infrastructure powered by clean solar; optimize commuter bus transit routes to maximize vehicle occupancy.

Target KPI: Fleet electrification percentage; average commuting passenger load

Expected Impact: Estimated 20%–30% mitigation in transport footprint

4. Scope 3 – Paper & Administrative Supplies

Low Priority | Short-term (0–3 months)

Evidence from Assessment: Paper consumption contributes 3.92 t CO_2e/yr.

Recommended Action: Institutionalize a paperless campus protocol by digitizing academic submissions, examination evaluations, and administrative approvals via an integrated ERP / LMS ecosystem.

Target KPI: Total reams of A4 paper procured per academic semester

Expected Impact: Estimated 60%–80% curtailment in physical paper consumption

5. Biological Carbon Sinks & Green Cover

Medium Priority | Continuous (Annual Drive)

Evidence from Assessment: Existing vegetative cover sequesters 3872.2 kg CO_2e/yr (160 cataloged trees).

Recommended Action: Initiate targeted afforestation of high-sequestration indigenous species (such as Peepal, Banyan, Neem, and Tamarind) across campus perimeter zones, creating high-density micro-forests (Miyawaki method).

Target KPI: Net tree survival count and canopy cover index

Expected Impact: Increases annual biological sequestration by 20–25 kg CO_2/tree/year

Strategic Decarbonization Roadmap & Phased Milestones

Achieving substantive emission reductions requires a structured, multi-phase operational roadmap that synchronizes capital budgeting with academic scheduling.

Phase 1: Operational Efficiency & Immediate Interventions (Months 0 – 6)

- Enact campus-wide paperless examination and administrative submission protocols.
- Pilot commercial induction cooking ranges in hostel dining kitchens to displace peak LPG cylinder usage.
- Complete LED lighting and BLDC ceiling fan retrofits across high-occupancy lecture blocks.
- Conduct comprehensive thermographic audits of power distribution panels to curtail transmission losses.

Phase 2: Clean Technology & Supply Integration (Months 6 – 18)

- Expand rooftop solar PV capacity to satisfy up to 50% of diurnal instructional electricity requirements.
- Deploy institutional electric buggies for inter-departmental mobility and cargo logistics.
- Install automated energy sub-metering IoT hubs with real-time departmental carbon tracking dashboards.
- Construct a dedicated wet organic waste anaerobic biomethanation plant.

Phase 3: Net-Zero Campus Leadership & Carbon Neutrality (Months 18 – 36)

- Establish battery energy storage systems (BESS) to buffer nighttime hostel power loads with clean daytime solar.
- Expand campus perimeter native tree density through the Miyawaki afforestation method, doubling biological sequestration.
- Seek formal third-party accredited ISO 14064-1 verification for NAAC, NIRF, and international sustainability ranking.

17. Conclusion

The greenhouse gas baseline established for Shri Hiralal Hastimal (Jain Brothers, Jalgaon) Polytechnic provides a transparent, scientifically defensible platform for institutional climate leadership. With a net operational carbon footprint of 44.03 t CO_2e/year (44,034.78 kg CO_2e) and an existing offset contribution of 6.63 t CO_2e/year, the campus is well-positioned to achieve accelerated decarbonization. Implementing the targeted engineering recommendations—namely kitchen thermal electrification, solar PV scaling, transit optimization, and continuous native afforestation—will steer the campus toward meaningful Net-Zero alignment and enhanced accreditation standing.

18. References

1. **World Resources Institute (WRI) & World Business Council for Sustainable Development (WBCSD).** "The Greenhouse Gas Protocol: A Corporate Accounting and Reporting Standard (Revised Edition)."
2. **International Organization for Standardization (ISO).** "ISO 14064-1:2018 Specification with guidance at the organization level for quantification and reporting of greenhouse gas emissions and removals."
3. **Central Electricity Authority (CEA), Ministry of Power, Government of India.** "CO_2 Baseline Database for the Indian Power Sector," User Guide Version 19.0 (Grid Emission Factor: 0.792 kg CO_2e/kWh).
4. **Intergovernmental Panel on Climate Change (IPCC).** "2006 IPCC Guidelines for National Greenhouse Gas Inventories & 2019 Refinement."
5. **UK Department for Environment, Food & Rural Affairs (DEFRA).** "GHG Conversion Factors for Company Reporting," 2024/2025.
6. **Bureau of Energy Efficiency (BEE), Ministry of Power, India.** "Energy Conservation Building Code (ECBC) for Educational Institutions."

Annexure-1

Details of Carbon Footprint Estimation

Name of Institute / Organization:	**Shri Hiralal Hastimal (Jain Brothers, Jalgaon) Polytechnic**
Campus Postal Address:	SNJB Jain Gurukul Neminagar, Chandwad, Nashik - 423101, Maharashtra
Official Contact Details:	+91 2556-252127
Official Email Address:	principalpoly@snjb.org
Institutional Website:	https://www.snjb.org
Total Campus Land Area:	4 acres (≈ 16,187.4 m²)
Total Built-up Facility Area:	16,187.4 m²
Geographic Coordinates:	Latitude: 20.33° N • Longitude: 74.23° E
Campus Elevation Above MSL:	585 m
Active Population Census:	1,420 occupants (1,300 students, 120 faculty & staff)
Assessment Reporting Year:	Financial Year 2025–2026 (FY 2026)

Audit Methodology Note: All institutional metadata recorded above has been formally cross-referenced against campus estate registers, municipal building sanctioned plans, and administrative HR rolls to ensure complete baseline veracity.

Annexure-1 (Contd.)

GREEN COVER — Tree Species Biomass Sequestration Ledger

Carbon sequestration quantification is based on species-specific biomass growth equations and annual carbon assimilation coefficients codified by the Indian Council of Forestry Research and Education (ICFRE).

Sr.	Botanical & Common Species Name	Tree Count	Absorption Factor (kg/tree/yr)	Total Removal (kg CO_2/yr)
1	Neem Tree (Azadirachta indica)	**110**	21.77	**2394.7**
2	Peepal Tree (Ficus religiosa)	**35**	28.50	**997.5**
3	Banyan Tree (Ficus benghalensis)	**15**	32.00	**480.0**
CUMULATIVE CAMPUS GREEN COVER TOTALS:		**160**	**—**	**3872.2**

Annexure-2

Calculation Methodologies, Formulas & Conversion Factors

Formula A1: Diesel Generator Combustion (Scope 1)

$E_DG\ (kg\ CO_2e) = Volume_Diesel\ (Liters/year) \times 2.684\ kg\ CO_2e/L$

Source: IPCC 2006 Guidelines for National Greenhouse Gas Inventories, Energy Volume.

Formula A2: Liquefied Petroleum Gas (LPG) Thermal Combustion (Scope 1)

$E_LPG\ (kg\ CO_2e) = Mass_LPG\ (kg/year) \times 2.988\ kg\ CO_2e/kg$

Source: UK DEFRA / IPCC 2006 Stationary Combustion coefficients for commercial LPG.

Formula A3: Purchased Grid Electricity Consumption (Scope 2)

$E_Grid\ (kg\ CO_2e) = Consumption\ (kWh/year) \times 0.792\ kg\ CO_2e/kWh$

Source: Central Electricity Authority (CEA), CO_2 Baseline Database for the Indian Power Sector, User Guide Version 19.0.

Formula A4: Commuting Vehicular Transportation (Scope 3)

$E_Transit\ (kg\ CO_2e) = Count \times Daily\ Distance\ (km) \times Emission\ Factor \times 0.0485 \times 240\ academic\ days$

Source: Automotive Research Association of India (ARAI) emission standards.

Formula A5: Paper Procurement Carbon Footprint (Scope 3)

$E_Paper\ (kg\ CO_2e) = Paper\ Mass\ (kg/year) \times 3.591\ kg\ CO_2e/kg$

Source: Environmental Paper Network (EPN) lifecycle cradle-to-gate inventory.

Formula A6: Solar Photovoltaic Grid Avoidance Offset

$Offset_Solar\ (kg\ CO_2e) = Solar\ Generation\ (kWh/year) \times 0.792\ kg\ CO_2e/kWh$

Displaces marginal fossil generation from the national grid at identical CEA factor.

Annexure-2 (Contd.)

Arithmetic Verification Proofs & Balance Verification

To ensure total mathematical transparency under ISO 14064-1:2018 requirements, the step-by-step arithmetic aggregation connecting primary inputs to final certified numbers is documented below:

Step	Arithmetic Equation	Result (kg CO_2e)	Metric Tons
Step 1	Scope 1 Subtotal (DG + LPG + Fleet)	26480.01	26.480
Step 2	Scope 2 Subtotal (Purchased Electricity)	3129.98	3.130
Step 3	Scope 3 Subtotal (Commuting + Paper)	21050.92	21.051
Step 4	**Gross Footprint = Step 1 + Step 2 + Step 3**	**50660.92**	**50.661**
Step 5	Verified Offsets (Solar + Trees + Waste)	6626.14	6.626
Step 6	**NET OPERATIONAL FOOTPRINT (Step 4 – Step 5)**	**44034.78**	**44.035**

Per Capita Integrity Proof:

Intensity = 44034.78 kg CO_2e ÷ 1420 occupants = **31.01 kg CO_2e/person/year**

Annexure-3

Data Collection Sheets & Audit Verification Sign-Off

The activity data used throughout this assessment was compiled from official institutional documents across the reporting period. All relevant meter readings, purchase invoices, and operational manifests were inspected and reconciled prior to computation.

Audited & Verified For: Shri Hiralal Hastimal (Jain Brothers, Jalgaon) Polytechnic

Auditing Body: CarbonTrack Sustainability Assessment & Climate Accreditation Cell

Verification Standards: ISO 14064-1:2018 • ISO 14064-3:2019 • GHG Protocol Corporate Standard

Document Authentication Code: CT-SUST-GHG-2026-1172

Date of Final Certification: October 15, 2026

Environmental Auditor Declaration

We hereby confirm that the greenhouse gas assertion of **Shri Hiralal Hastimal (Jain Brothers, Jalgaon) Polytechnic** for Financial Year 2025–2026 (FY 2026) has been audited in accordance with the specifications of ISO 14064-3. Based on our evaluation of the data and evidence provided, nothing has come to our attention that causes us to believe that the GHG inventory is not fairly stated in all material respects in accordance with ISO 14064-1:2018.

Certified Lead Environmental Auditor
CarbonTrack Sustainability Assessment Cell

Authorized Signatory / Registrar
Shri Hiralal Hastimal (Jain Brothers, Jalgaon) Polytechnic